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Ahmedabad University

CSE 516 – Probabilistic Graphical Models

Milestone 2: Technical Foundation

Bayesian Personalized Recommendation using Latent Trait Models & Expectation Propagation

Category: Theoretical AI

Group 4

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GitHub Repository

https://github.com/Jyotier2006/M2_g4_th_apl_recommend-sys_team

Submission Date: 18 October 2026

Abstract

We implement a static, identity-only Matchbox model to study the trade-off between single-pass assumed-density filtering (ADF) and repeated expectation propagation (EP). Gaussian latent traits and biases generate ordinal ratings through fixed thresholds; learning uses hybrid EP/variational message passing (VMP). Across three MovieLens 1M splits, eight-trait EP reduces test RMSE from 0.8980 ± 0.0049 to 0.8633 ± 0.0040, while fitting time rises from 2.64 to 13.63 seconds. Conventional matrix factorization achieves a close RMSE of 0.8646 ± 0.0031 at lower cost. Nine validation checks pass, but a numerical reference exposes approximation error and calibration does not improve. These initial results establish the M2 technical foundation, not a full Matchbox reproduction or SOTA claim.

2 Problem and progress from M1

Sparse user–item ratings leave uncertainty about both preferences and item characteristics. A latent-trait graphical model shares evidence across ratings through common user and item parameters, while Bayesian learning retains uncertainty about those parameters. Our theoretical question is: *how do ADF and repeated EP differ in predictive quality, uncertainty, computation and memory?*

M0 selected Matchbox and MovieLens 1M; M1 defined the model, baselines and experimental plan. M2 implements the graph, parameter updates, prediction, baselines, validation and first benchmark runs. The graph structure is fixed: we learn parameter distributions, not edges. The present scope uses static ID features and fixed global rating thresholds. Metadata, learned user thresholds, controlled cold-start and wider sensitivity studies remain M3 work.

3 Probabilistic representation

For user u and item i, let $s_u, t_i \in \mathbb{R}^K$ denote latent traits and b_u, b_i denote biases. For each observed pair $(u, i) \in \mathcal{D}$,

$$z_{uik} = s_{uk} t_{ik}, \qquad a_{ui} = \mu + b_u + b_i + \sum_{k=1}^{K} z_{uik}, \tag{1}$$

$$y_{ui} \mid a_{ui} \sim \mathcal{N}(a_{ui}, \beta^2), \qquad R_{ui} = r \iff c_{r-1} < y_{ui} \le c_r. \tag{2}$$

Only the rating is observed. The cuts are $c = (-\infty, 1.5, 2.5, 3.5, 4.5, +\infty)$; μ is the training mean, treated as fixed. Independent priors are

$$s_{uk} \sim \mathcal{N}(m^0_{uk}, 0.25), \quad t_{ik} \sim \mathcal{N}(n^0_{ik}, 0.25), \quad b_u, b_i \sim \mathcal{N}(0, 1). \tag{3}$$

Trait prior means are sampled once from $\mathcal{N}(0, 0.1^2)$ using the run seed. This breaks bilinear sign symmetry without using held-out ratings.

With $\Theta = (S, T, b_U, b_I)$, the joint factorization is

$$\begin{aligned} p(\Theta, Z, A, Y, R) = p(\Theta) \prod_{(u,i)\in\mathcal{D}} & \left[\prod_k \delta(z_{uik} - s_{uk} t_{ik}) \right] \\ & \times \delta\left(a_{ui} - \mu - b_u - b_i - \sum_k z_{uik} \right) \\ & \times \mathcal{N}(y_{ui}; a_{ui}, \beta^2) \mathbf{1}\{c_{R_{ui}-1} < y_{ui} \le c_{R_{ui}}\}. \end{aligned} \tag{4}$$

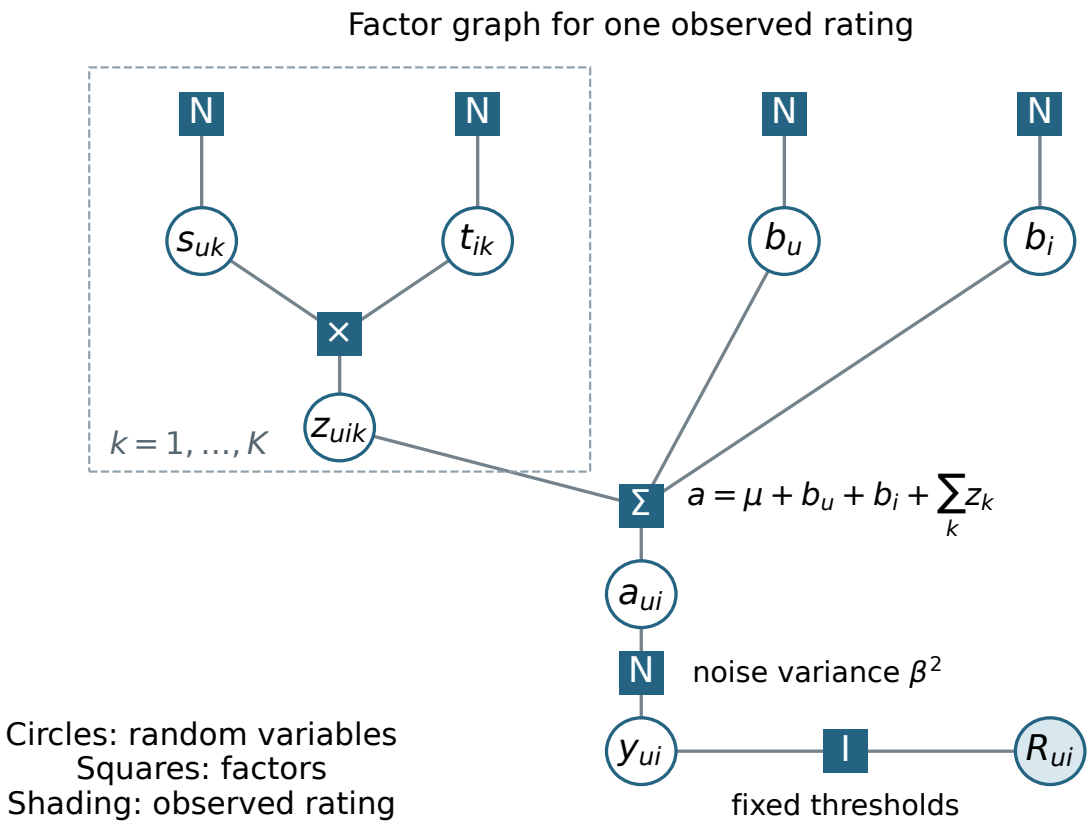


Figure 1: Factor graph for one rating. The plate repeats the trait product; user/item variables are shared across all corresponding ratings. Edges connect variables to the factors that depend on them.

Prior factors regularize parameters; products express compatibility; the sum adds biases; Gaussian noise captures residual variation; and the interval factor converts a continuous score into a rating. The code also exports a concrete variable/factor/edge graph in `graph_instance.json`.

Ratings are conditionally independent given all relevant parameters. Ratings sharing a user or item are generally dependent after marginalization. Independent priors do not imply independent posterior parameters: the factorized Gaussian posterior is an approximation. The model assumes stationary preferences and does not model why a movie was rated or identify causal effects.

4 Learning and inference

4.1 Posterior learning and prediction query

Training approximates $p(\Theta \mid R_{\mathcal{D}})$ by a product of univariate Gaussians, storing precision $\lambda = 1/v$ and weighted mean $\eta = m/v$. Thresholds, noise and prior variances remain fixed. Prediction computes $p(R_{ui} = r \mid R_{\mathcal{D}})$ for an unobserved pair.

Under the factorized approximation, product moments are

$$M_k = \mathbb{E}[s_k t_k] = m_{s_k} m_{t_k}, \tag{5}$$

$$V_k = \mathrm{Var}(s_k t_k) = v_{s_k} v_{t_k} + v_{s_k} m_{t_k}^2 + v_{t_k} m_{s_k}^2. \tag{6}$$

Set $m_y = \mu + m_{b_u} + m_{b_i} + \sum_k M_k$ and $v_y = \beta^2 + v_{b_u} + v_{b_i} + \sum_k V_k$. Approximating the score as Gaussian gives

$$\widehat{p}_r = \Phi\left(\frac{c_r - m_y}{\sqrt{v_y}}\right) - \Phi\left(\frac{c_{r-1} - m_y}{\sqrt{v_y}}\right), \qquad \widehat{R} = \sum_{r=1}^{5} r\widehat{p}_r. \tag{7}$$

The moments are exact under the factorized q; the Gaussian score approximation is additional. Probabilities use a 10^{-15} numerical floor and renormalization.

4.2 Local hybrid EP/VMP update

Exact posterior inference is intractable at this scale because bilinear products and ordinal observations couple many parameters. Following Matchbox [4], we use VMP at products and EP or exact Gaussian messages elsewhere. A local loop alternates forward moments, ordinal moment matching and reverse messages, reconstructing marginals from a fixed cavity and the current message.

For an incoming score $\mathcal{N}(m, v)$, define $\alpha = (c_{r-1} - m)/\sqrt{v}$, $\gamma = (c_r - m)/\sqrt{v}$ and $Z = \Phi(\gamma) - \Phi(\alpha)$. Truncated moments are

$$\widehat{m} = m + \sqrt{v}\,\frac{\phi(\alpha) - \phi(\gamma)}{Z}, \tag{8}$$

$$\widehat{v} = v\left[1 + \frac{\alpha\phi(\alpha) - \gamma\phi(\gamma)}{Z} - \left(\frac{\phi(\alpha) - \phi(\gamma)}{Z}\right)^2\right]. \tag{9}$$

Gaussian division yields $\lambda_o = \widehat{v}^{-1} - v^{-1}$ and $\eta_o = \widehat{m}/\widehat{v} - m/v$. Write the outgoing Gaussian as $\mathcal{N}(y; m_o, v_o)$. Reverse propagation to product z_k gives

$$d_k = m_o - \mu - m_{b_u}^{\backslash n} - m_{b_i}^{\backslash n} - \sum_{\ell \neq k} M_\ell, \tag{10}$$

$$w_k = v_o + \beta^2 + v_{b_u}^{\backslash n} + v_{b_i}^{\backslash n} + \sum_{\ell \neq k} V_\ell. \tag{11}$$

The VMP message to s_k is

$$\lambda_{f \to s_k} = \frac{\mathbb{E}[t_k^2]}{w_k}, \qquad \eta_{f \to s_k} = \frac{d_k \mathbb{E}[t_k]}{w_k}, \quad \mathbb{E}[t_k^2] = v_{t_k} + m_{t_k}^2. \tag{12}$$

The item update is symmetric. A bias receives the analogous Gaussian sum message with its own cavity term omitted. Product moments use current local marginals; bias inputs remain cavities. Local damping is 0.7, with at most 10 iterations and an absolute mean/variance change tolerance of 10^{-4}.

4.3 ADF versus repeated EP

EP caches a Gaussian site per rating. For each touched scalar parameter,

$$\lambda = \lambda_0 + \sum_n \lambda_n, \quad \eta = \eta_0 + \sum_n \eta_n; \qquad \lambda^{\backslash n} = \lambda - \lambda_n, \quad \eta^{\backslash n} = \eta - \eta_n. \tag{13}$$

An EP pass visits the fixed training order, subtracts the old site's natural parameters, runs the local update, and replaces that site. The replacement is damped with weight 1 on pass one and 0.5 thereafter.

ADF processes each rating once and discards its site. Thus EP's first pass matches ADF under identical priors and order. Replaying ADF without cavity removal would double-count evidence. Our ADF interface rejects a second pass. Five EP passes are a computational budget, not proof of convergence.

5 Mathematical validation and complexity

Nine checks pass: truncated moments against quadrature; exact marginals for a single $K = 0$ ordinal factor; ADF–EP1 equality; the prior-plus-sites invariant; no double counting on a single-rating revisit; product moments against Monte Carlo; probability/metric sanity; disjoint reproducible splits; and checkpoint reload. These validate specific formulas and invariants, not exactness of the whole approximation.

A separate bilinear test uses $R = 5$, offset 3, noise variance 1, $s \sim \mathcal{N}(0.5, 0.4)$ and $t \sim \mathcal{N}(0.8, 0.4)$, with bias variances 10^{-8}. Tensor Gauss–Hermite quadrature integrates the likelihood $P(y > 4.5 \mid s, t)$, analytically integrating the biases. Increasing the order from 40 to 80 changes the reference moments by at most 1.04×10^{-10}. For this diagnostic only, the local cap is 500 and tolerance 10^{-12}; it converges after 50 iterations.

Quantity	Quadrature	Hybrid	Absolute error
$\mathbb{E}[s]$	0.922831	0.933888	0.011056
$\mathrm{Var}(s)$	0.325234	0.292029	0.033206
$\mathbb{E}[t]$	1.123960	1.136053	0.012093
$\mathrm{Var}(t)$	0.351319	0.315457	0.035862
$\mathrm{Cov}(s, t)$	0.070526	0.000000	0.070526

Table 1: One-observation validation against 80-point-per-axis quadrature. The hybrid approximation cannot retain the nonzero posterior covariance.

Means differ by less than 0.013, but both variances are about 10% too small. This motivates checking uncertainty as well as rating error.

For N ratings, K traits, L local iterations and T passes, training costs $O(TNL(K + 1))$ and batch prediction $O(N(K + 1))$. Posterior storage is $O((U + I)(K + 1))$. Repeated EP additionally stores

$$8 \times 2 \times N(2K + 2) = 32N(K + 1) \text{ bytes} \tag{14}$$

in 64-bit rating sites. At $N = 800{,}167$, $K = 8$, this is 219.77 MiB; ADF has no site cache. The 87,714 scalar parameters require 175,428 natural parameters (1.34 MiB). These array sizes exclude data, temporaries and process overhead.

6 Benchmark setup and baselines

6.1 Dataset and fair comparison

MovieLens 1M [1, 2] contains 1,000,209 ratings from 6,040 users on 3,706 rated movies; the metadata catalogue contains 3,883 movies. The archive and input files are SHA-256 checked. No duplicate user–item pair exists. Rating density is 4.468%; missing entries are unobserved, not zero ratings.

Seeds 42, 43 and 44 each produce 800,167 training, 100,020 validation and 100,022 test rows. Every method uses the same split for a seed. ID mapping uses IDs only; fitted statistics use training ratings. ADF and EP share the training order, and EP retains it across passes. The test sets have no unseen users; 26, 9 and 13 ratings involve unseen items and use prior information. This is a random-interaction benchmark, not a chronological or controlled cold-start test.

6.2 Implemented baselines and settings

Mean baselines. Global rating frequencies are g_r. User/item class counts n_r produce smoothed probabilities $(n_r + 10g_r)/(n + 10)$, whose expectation is a shrinkage mean. Unseen entities use the global distribution.

Biased matrix factorization (MF). The predictor is $\mu + b_u + b_i + p_u^T q_i$, fitted by squared-error SGD with $K = 8$, learning rate 0.007, epoch decay 0.98, regularization 0.03 and 25 epochs. Validation clipped-point RMSE selects the checkpoint (epoch 25 in all runs). A Gaussian scale is then fitted using validation NLL within standard deviations $[0.2, 3]$; fixed ordinal cuts yield probabilities. We report both their expectation and the conventional clipped point prediction. The wrapper models residual spread, not parameter uncertainty.

ADF and EP. Both use the same priors, local update and noise variance $\beta^2 = 1$. Trait counts are $K = 2, 8$; EP checkpoints are passes 1, 3 and 5. Settings were fixed before the full reference test run. An earlier 100,000-row development run checked execution and is excluded from benchmark results.

6.3 KPIs, variation and reproducibility

MAE and RMSE use $\widehat{R}$ unless labelled clipped point. NLL is mean $-\log \widehat{p}_R$ in nats; Brier is mean summed squared error over the five classes; top-class ECE uses 10 equal-width confidence bins. We also record 90% equal-tail rating intervals, categorical-median MAE, time and iteration diagnostics. Results are mean ± sample SD over three runs. Splits overlap, and seeds change both splitting and initialization, so no significance claim is made from these three runs.

Experiments ran on 17 October 2026 with Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0 and Numba 0.68.0 on Linux x86-64. Kernels are sequential and numerical thread settings equal one. Fit timing excludes compilation, I/O and KPI evaluation; MF includes validation checkpointing and scale fitting. Prediction is timed separately. Results are hardware-dependent wall times.

The accompanying package provides source code, the official data downloader, configuration, split indices, seeds, logs, environment details and result-generation scripts. The Code README documents validation, checkpoint prediction and full reproduction; raw MovieLens files are not redistributed.

7 Initial results and interpretation

7.1 Prediction and baseline comparison

EP5 improves mean RMSE by 0.0347 (3.86%) and NLL by 0.0344 over ADF in all three runs. Revisiting ratings can use later evidence without double counting; this supports the expected accuracy/cost trade-off.

Method	MAE ↓	RMSE ↓	NLL ↓
Global mean	0.9342 ± 0.0018	1.1174 ± 0.0022	1.4558 ± 0.0018
User mean	0.8313 ± 0.0019	1.0353 ± 0.0020	1.3469 ± 0.0014
Item mean	0.7827 ± 0.0029	0.9785 ± 0.0037	1.3205 ± 0.0039
MF	0.6907 ± 0.0029	0.8681 ± 0.0031	1.2210 ± 0.0033
ADF	0.7101 ± 0.0039	0.8980 ± 0.0049	1.2520 ± 0.0046
EP3	0.6887 ± 0.0034	0.8731 ± 0.0046	1.2271 ± 0.0039
EP5	0.6800 ± 0.0029	0.8633 ± 0.0040	1.2176 ± 0.0032
MF, clipped	0.6809 ± 0.0029	0.8646 ± 0.0031	same model

Table 2: Test results, mean ± sample SD over seeds 42–44. All latent models use $K = 8$. Main rows use the categorical mean; the final row shows conventional clipped MF. Both MF decoders share the same probability model and NLL.

The mean RMSE advantage over conventional clipped MF is only 0.0013, and MF wins on seed 43. EP5 is therefore competitive in this experiment, not clearly superior. MF also fits faster. Mean predictors confirm that personalization improves over aggregate statistics.

7.2 Iteration, dimension and computational cost

Method	Fit time (s)	Site memory (MiB)
MF, $K = 8$	3.000 ± 0.565	0.00
ADF, $K = 8$	2.644 ± 0.212	0.00
EP3, $K = 8$	8.187 ± 0.421	219.77
EP5, $K = 8$	13.632 ± 0.879	219.77

Table 3: Fitting time, mean ± sample SD; site-array memory is deterministic. Prediction for EP5 at $K = 8$ takes about 0.56 microseconds per rating in a batch.

At $K = 8$, EP5 costs about 5.16 times ADF's fitting time. Eight traits improve EP5 RMSE over two (0.8633 versus 0.8760), but triple site storage from 73.26 to 219.77 MiB. Two dimensions do not establish an optimal model capacity.

No numerical update was skipped. At most 1,027 of 800,167 local updates (0.128%) hit the iteration cap. By pass five at $K = 8$, this falls to 10–17 per seed, but RMS changes in parameter means remain 0.0328–0.0347. Local stability and lower validation error therefore do not establish outer convergence.

7.3 Uncertainty and calibration

At $K = 8$, Brier improves from 0.6681 to 0.6547, but ECE worsens from 0.0449 to 0.0562. Better NLL and RMSE do not guarantee better calibration. The reliability curve shows underconfident top-class predictions in substantial probability regions. EP5's nominal 90% intervals cover 98.22% of ratings and span 3.82 categories on average; discretization and fixed noise make these broad, and high coverage alone does not establish calibration.

The small quadrature test concerns uncertainty in latent parameters; rating intervals also contain observation noise and discretization. Thus local variance underestimation and broad predictive intervals are not contradictory.

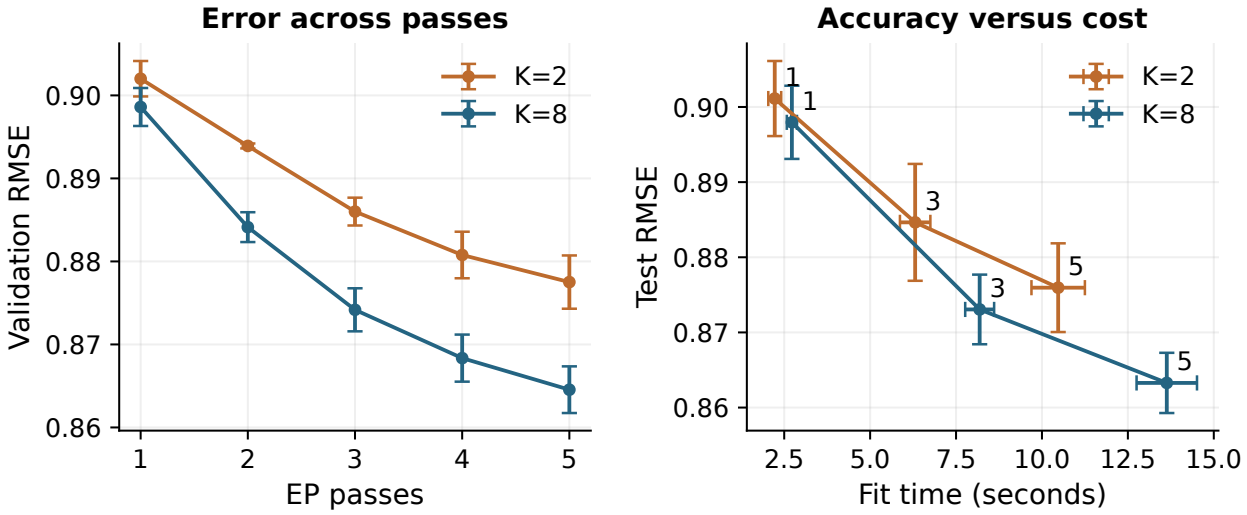


Figure 2: Validation error by pass and test error versus cumulative fitting time. Error bars show sample SD; right-panel labels are the EP pass counts.

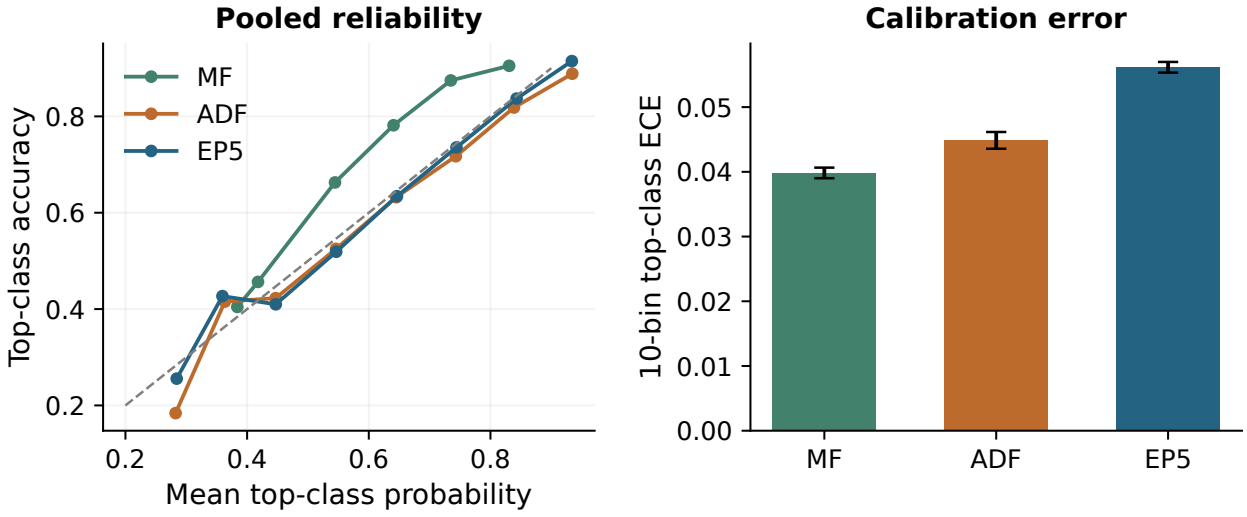


Figure 3: Calibration evidence. Pooled bins are descriptive because splits overlap. Bins with at most 50 pooled cases are omitted only from the curve; ECE uses all bins.

8 Related work and revised SOTA positioning

Matchbox: Microsoft Research, WWW 2009. Stern, Herbrich and Graepel [4] combine latent features, optional metadata, feedback models and hybrid message passing. ADF provides efficient online Bayesian updates; repeated EP revisits cached rating contributions. Their experiments study recommendation accuracy, cold-start behaviour and scalability. Our implementation retains the inference question but uses static ID features, fixed cuts and its own schedule, rather than reproducing Infer.NET or the full system. Published scores use different protocols and are not directly comparable.

PreRec: AWS AI Labs affiliation, WSDM 2024. Lin et al. [3] use a hierarchical Bayesian model with neural user/item encoders, domain variables and popularity effects. Training is MAP optimization with Adam; zero-shot scoring changes domain and popularity mechanisms. XMarket/Online Retail experiments evaluate cross-domain sequential ranking and limited-data adaptation. Reported gains support modelling transfer and bias in that setting, subject to the

learned representations and assumed bias mechanisms. They do not establish an EP advantage or provide a MovieLens rating baseline.

Since M1, our position has become more specific: M2 supplies a reproducible comparison of approximate inference and exposes uncertainty limitations. It does not demonstrate a new algorithm, causal debiasing or cross-domain transfer. PreRec remains a conceptual anchor for future modelling choices, not an executed competitor or an exhaustive survey of current recommendation research.

9 Limitations, M3 plan and conclusion

The main limits are factorized posteriors, approximate product messages, fixed thresholds/noise, no metadata, order-dependent ADF and only three overlapping splits. The MF and PGM models have not received matched exhaustive tuning; random-interaction scores may be optimistic for future or new-user deployment.

M3 will study prior/noise/damping/order sensitivity, longer EP runs with stopping diagnostics, metadata and learned-threshold ablations, controlled user cold-start and reduced density, and uncertainty analysis accounting for user clustering.

M2 now connects representation, Bayesian parameter learning, prediction, baselines and measurable KPIs. Repeated EP improves on ADF at a clear time and memory cost, while remaining close to conventional MF. Its calibration and approximation errors give concrete questions for the next milestone.

References

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